INCH-POUND

MIL-DTL-83513/3H
22 March 2007
SUPERSEDING
MIL-DTL-83513/3G
25 June 2002

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, PIN CONTACTS, CLASS M, CRIMP TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83513.

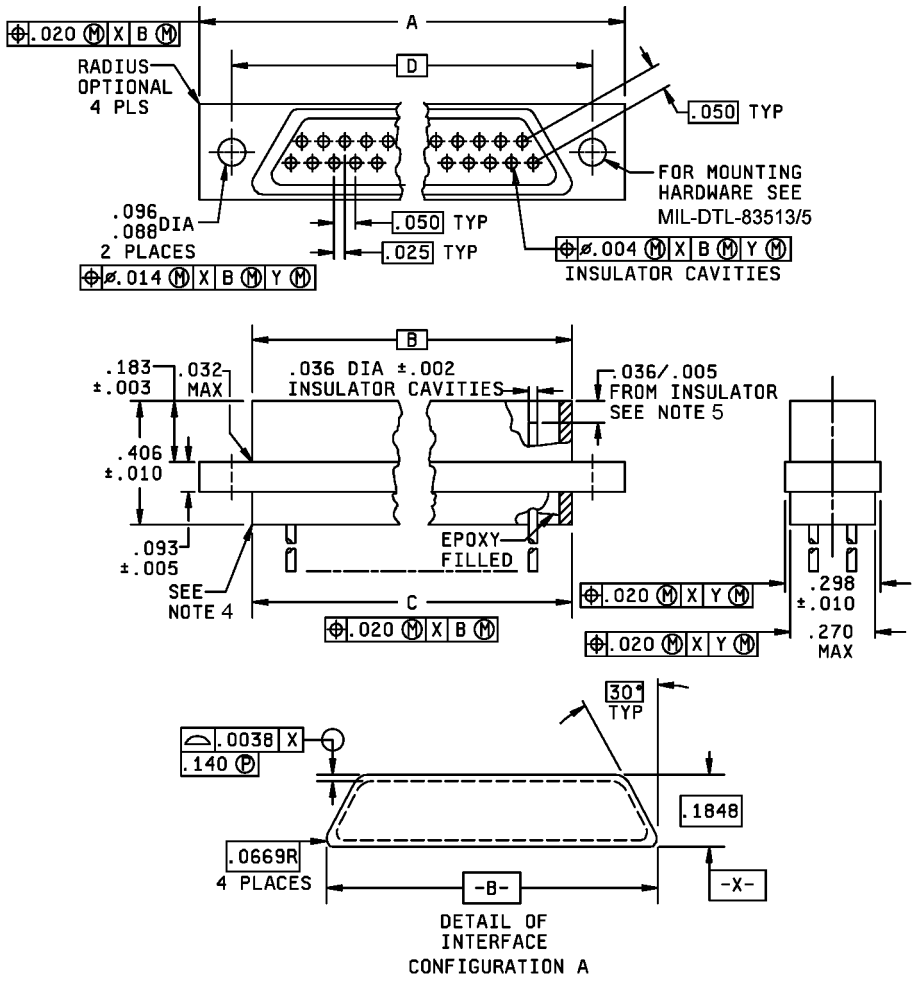


FIGURE 1. Connector, plug, .050 spacing.

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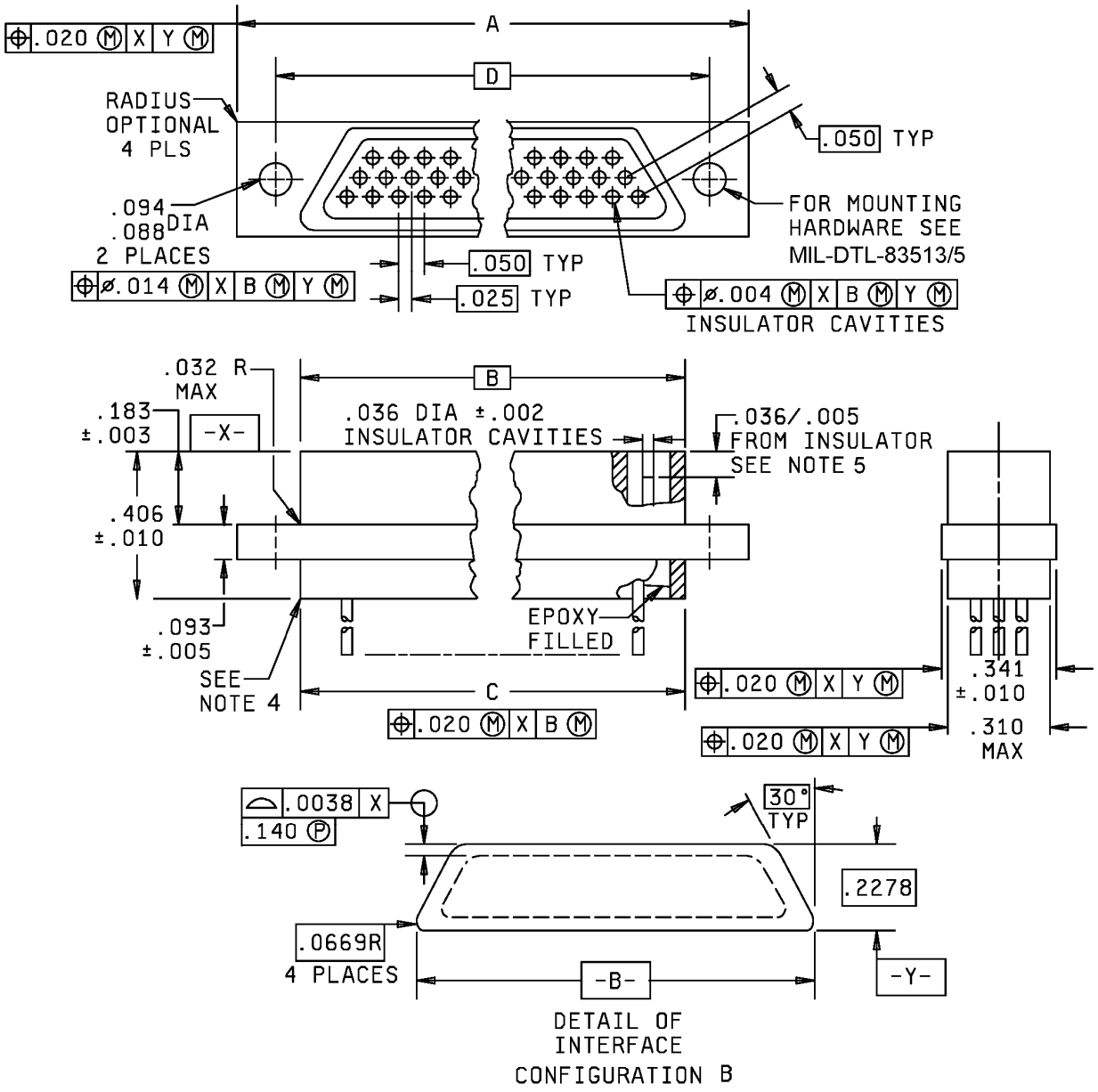


FIGURE 1. Connector, plug, .050 spacing - Continued.

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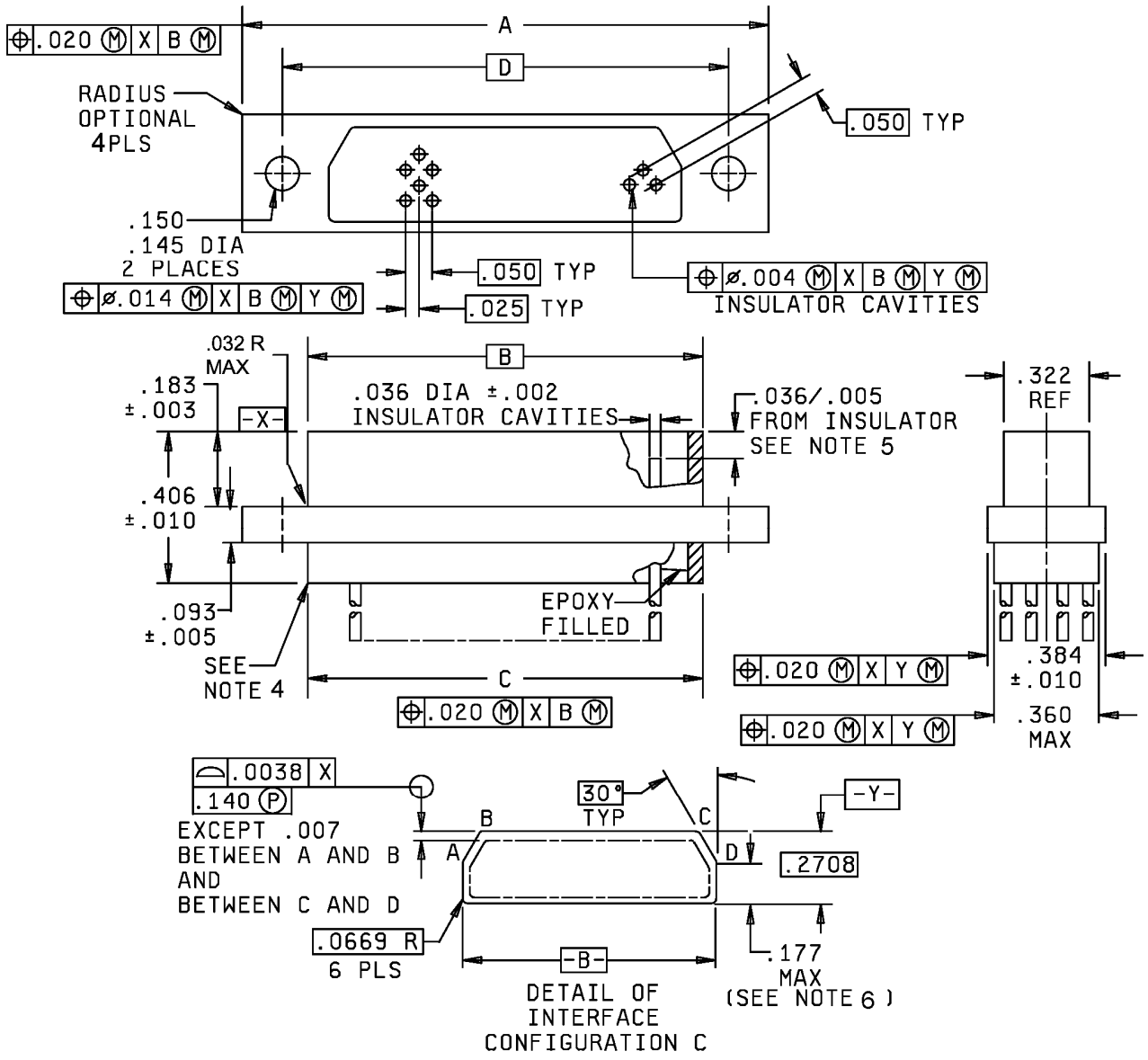


FIGURE 1. Connector, plug, .050 spacing - Continued.

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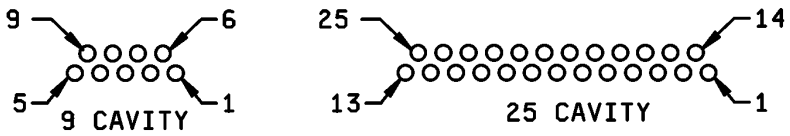
Number of cavities	A .010 (.25)	B BSC	C +.010 -.018 +(.25) -(.46)	D BSC	Configuration letter
9	.775 (19.68)	.3338 (8.48)	.390 (9.91)	.565 (14.35)	A
15	.925 (23.50)	.4838 (12.29)	.540 (13.72)	.715 (18.16)	A
21	1.075 (27.30)	.6338 (16.10)	.690 (17.53)	.865 (21.97)	A
25	1.175 (29.84)	.7338 (18.64)	.790 (20.07)	.965 (24.51)	A
31	1.325 (33.66)	.8838 (22.45)	.940 (23.88)	1.115 (28.32)	A
37	1.475 (37.46)	1.0338 (26.26)	1.090 (27.69)	1.265 (32.13)	A
51	1.425 (36.20)	.9838 (24.99)	1.040 (26.42)	1.215 (30.86)	B
100	2.160 (54.86)	1.3838 (35.15)	1.432 (36.37)	1.800 (45.72)	C

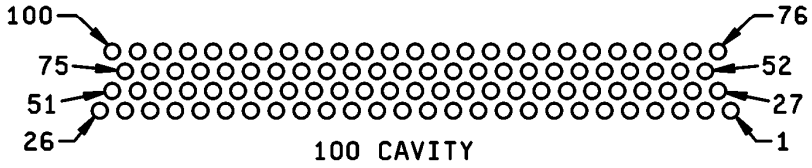
Inches	mm	Inches	mm
.0015	0.038	.094	2.39
.002	0.05	.1139	2.893
.003	0.08	.117	2.97
.0038	0.097	.123	3.12
.004	0.10	.145	3.68
.005	0.13	.150	3.81
.007	0.18	.173	4.39
.010	0.25	.181	4.60
.014	0.36	.182	4.62
.020	0.51	.183	4.65
.022	0.56	.224	5.69
.025	0.64	.2278	5.786
.036	0.91	.267	6.78
.050	1.27	.270	6.86
.0669	1.699	.298	7.57
.0671	1.704	.310	7.87
.089	2.26	.341	8.66
.090	2.29	.360	9.14
.0924	2.345	.384	9.75
.093	2.36	.406	10.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4 Point from which the lead length is measured.
5. Shell shall be flush to insulator within ± .005 (0.13 mm).
6. Dimension line at point D locates the point of intersection of the two adjacent side of the receptacle interface.

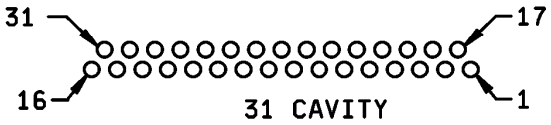
FIGURE 1. Connector, plug, .050 spacing - Continued.

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NOTES:
1. Engaging face of pin insert shown.
2. Cavity identification numbers are for reference only and do not appear on part.

FIGURE 2. Insert arrangement.

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REQUIREMENTS:

Dimensions and configurations: See figure 1. This specification sheet describes the pin side of a rectangular metal shell connector. This connector uses reverse gender contact, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Materials and finish:

Shell: The requirements for shell materials shall be in accordance with MIL-DTL-83513.

Contact identification: See figure 2 for contact layout identification.

Contact connection: The pin contact (being recessed in the insulator) is normally connected to the live side of the circuit.

Current rating, maximum: 3 amperes per contact.

Mounting and mating hardware: Mounting hardware is to be ordered separately. Insert arrangements A through G, are in accordance with MIL-DTL-83513/5, configurations A and B. Insert arrangement H, is in accordance with MIL-DTL-83513/5, configuration C.

Mating receptacle: Mating receptacle in accordance with MIL-DTL-83513/2 or MIL-DTL-83513/4.

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Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, a letter from the insert column and the shell finish.

M83513/03 - A 01 1/ C

M83513/03 - Specification sheet number

A - Insert arrangement (see figure 2)

A = 9
B = 15
C = 21
D = 25
E = 31
F = 37
G = 51
H = 100

01 1/ - Wire type

01 = M22759/11-26-9, 18 inches long 2/ 3/
02 = M22759/11-26-9, 36 inches long 2/ 3/
03 = M22759/11-26-( ), 18 inches long 3/ 4/ 5/
04 = M22759/11-26-( ), 36 inches long 3/ 4/ 5/
05 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 micro inches; 0.5 inch long 6/ 7/
06 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 micro inches; 1.0 inch long 6/ 7/
07 = A-A-59551, type S, 25 AWG, tin plated 100 micro inches; 0.5 inch long 6/ 7/ 9/
08 = A-A-59551, type S, 25 AWG, tin plated 100 micro inches; 1.0 inch long 6/ 7/ 9/
09 = M22759/33-26-9, 18 inches long 3/
10 = M22759/33-26-9, 36 inches long 3/
11 = M22759/33-26-( ), 18 inches long 3/ 4/
12 = M22759/33-26-( ), 36 inches long 3/ 4/
13 = M22759/11-26-9, 72 inches long 2/ 3/
14 = M22759/11-26-( ), 72 inches long 3/ 4/ 5/
15 = M22759/33-26-9, 72 inches long 3/
16 = M22759/33-26-( ), 72 inches long 3/ 4/
17 = 04035-26A-9, 18 inches long 2/ 3/
18 = 04035-26A-9, 36 inches long 2/ 3/
19 = 04035-26A-9, 72 inches long 2/ 3/
20 = 04035-26A-( ), 18 inches long 3/ 4/ 5/
21 = 04035-26A-( ), 36 inches long 3/ 4/ 5/
22 = 04035-26A-( ), 72 inches long 3/ 4/ 5/

C - Shell finish (Interface critical)

C = Cadmium
N = electroless nickel (shall be used for space applications)
P = Passivated stainless steel

1/ These connectors have leads attached of which wire type, wire size, wire color, and lead length are specified in PIN.
2/ For connectors intended for space applications, wire shall be M22759/33-26-9.
3/ Lead length tolerance +1.0, -0.0 inch.
4/ For connectors intended for space applications, wire shall be M22759/33-26-( ).
5/ Color coding in accordance with MIL-STD-681, system I, except the appropriate color code shall be the individual contact cavity number minus 1 for contact positions 1 through 10, repeat this color sequence for contact positions 11 through 20, etc. For example:

PIN	Identification
4	3
9	8
21	0
31	0
77	6
100	9

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6/ Lead length tolerance +0.2, -0.0 inch.
7/ Contact and wire may be welded or a single piece design.
8/ Corrosion has been experienced on M83513 connectors that are prewired with SAE AS22759/33 and stored in a sealed environment. CAUTION SHOULD BE EXCERCISED WHEN USING THIS WIRE.
9/ Pure tin. The use of pure tin is prohibited. Tin content used in connector materials shall not exceed 97 percent, and an alloy material shall be chosen to inhibit the growth of tin whiskers.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-83513, this document references the following:

A-A-59551
SAE AS22759/33
MIL-DTL-83513/2
MIL-DTL-83513/4
MIL-DTL-83513/5
MIL-STD-681

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4663-000)

Review activities:
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force – 99

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